

Network-1 Project Documentation

1. Network Design & Topology Overview

The network topology represents a Wide Area Network (WAN) interconnecting four distinct sites/routers:

- **Admin Center Router (Hub):** Acts as the central hub connecting to all other routers, providing central services like DHCP. It also connects to the ISP and performs NAT to the outside world.
- **Engineer Router:** Connects the Engineering department, which is segmented into multiple VLANs (Staff, Labs, and a Server network).
- **Science Router:** Connects the Science department LAN.
- **Art Router:** Connects the Art department LAN.
- **ISP Router:** Represents the Internet Service Provider, terminating the outside public connection.

This hub-and-spoke-like design uses Static Routing for inter-network reachability, with Site-to-Site VPN configured between the Engineer and Science branches for secure data transmission.

2. VLSM Addressing Table

Based on the configuration files, the IP addressing scheme utilizes Variable Length Subnet Masking (VLSM).

Location / Network	Network Address	Subnet Mask	First Usable IP	Last Usable IP	Broadcast Address	Default Gateway (Router IP)
Engineer - Labs (VLAN 20)	192.168.10.0	255.255.255.128 (/25)	192.168.10.1	192.168.10.126	192.168.10.127	192.168.10.1
Engineer - Staff (VLAN 10)	192.168.10.128	255.255.255.128 (/25)	192.168.10.129	192.168.10.254	192.168.10.255	192.168.10.129
Science LAN	192.168.11.0	255.255.255.128 (/25)	192.168.11.1	192.168.11.126	192.168.11.127	192.168.11.1
Art LAN	192.168.11.128	255.255.255.192 (/26)	192.168.11.129	192.168.11.190	192.168.11.191	192.168.11.129
Engineer - Servers (VLAN 30)	192.168.11.192	255.255.255.248 (/29)	192.168.11.193	192.168.11.198	192.168.11.199	192.168.11.193
WAN: Admin - Engineer	10.0.0.0	255.255.255.248 (/29)	10.0.0.1	10.0.0.6	10.0.0.7	N/A
WAN: Admin - Science	10.0.0.8	255.255.255.252 (/30)	10.0.0.9	10.0.0.10	10.0.0.11	N/A
WAN: Admin - Art	10.0.0.12	255.255.255.252 (/30)	10.0.0.13	10.0.0.14	10.0.0.15	N/A
WAN: Admin - ISP	200.0.0.0	255.255.255.248 (/29)	200.0.0.1	200.0.0.6	200.0.0.7	N/A

3. Protocol Implementation Details

DHCP (Dynamic Host Configuration Protocol)

The **Admin Router** acts as the central DHCP server for the entire network. Branch routers use the `ip helper-address` command on their LAN-facing interfaces to forward DHCP broadcasts to the Admin Router.

- **Pool 1an1v10:** Provides IPs for Engineer Staff VLAN 10 (192.168.10.128/25).
- **Pool 1an1v20:** Provides IPs for Engineer Labs VLAN 20 (192.168.10.0/25).
- **Pool 1an2:** Provides IPs for Science LAN (192.168.11.0/25).
- **Pool 1an3:** Provides IPs for Art LAN (192.168.11.128/26).

DNS & Web Server

- All DHCP clients are assigned 192.168.11.194 as their DNS Server. This server is located in the Engineer Server network (VLAN 30).
- Static NAT exposes the internal Web/DNS servers to the outside.
- **Note:** The packet tracer configuration contains the network layout for the DNS/Web servers but they are managed via the end-device interface directly rather than router CLI.

VLAN (Virtual LAN)

VLANs are implemented primarily in the Engineer department to segment broadcast domains:

- **VLAN 10:** Staff Network
- **VLAN 20:** Labs Network
- **VLAN 30:** Servers Network
- **Inter-VLAN Routing:** Router-on-a-Stick is configured on the Engineer Router using sub-interfaces (GigabitEthernet0/0/0.10, .20, .30) with encapsulation dot1Q.
- **Switches** (Switch_access_eng, Switch_core_eng, Switch_denstination_eng) are configured with appropriate Access ports for endpoints and Trunk ports for switch-to-switch and switch-to-router connections.

NAT (Network Address Translation)

NAT is configured on the **Admin Router** to translate private internal IPs to public IPs when traversing the ISP.

- **PAT (Port Address Translation / Overload):** `ip nat inside source list 1 interface Serial0/2/1 overload` allows all internal networks (permitted by ACL 1) to access the internet using the Admin Router's Serial0/2/1 IP address.
- **Static NAT:** `ip nat inside source static 192.168.11.194 200.0.0.3` and `192.168.11.195 200.0.0.4` map the internal DNS and Web servers to public IP addresses, making them accessible from the outside network.

VPN (Site-to-Site IPsec)

A Site-to-Site IPsec VPN tunnel is established between the **Engineer Router** and the **Science Router**.

- **ISAKMP Policy:** Group 2, AES Encryption, Pre-Shared Key (cisco).
- **Transform Set:** `ts using esp-aes and esp-sha-hmac`.
- **Crypto Map:** VPN is applied to the WAN-facing serial interfaces.
- **Interesting Traffic:** Defined by ACL 110. Traffic between the Engineer network (192.168.11.192/29) and Science network (192.168.11.0/25) triggers the encryption.

Static Routing

All routers use Static Routing (`ip route`) to reach remote networks.

- **Admin Router** has static routes pointing to the LAN networks behind the Engineer, Science, and Art routers.
- **Branch Routers** (Engineer, Science, Art) have static routes pointing back to the Admin Router for inter-branch reachability.

4. Device Configurations

Below are the complete `running-config` configurations for each networking device in the project.

Admin Router

```
!  
version 15.1  
hostname Router  
!  
ip dhcp pool lan1v10  
network 192.168.10.128 255.255.255.128  
default-router 192.168.10.129  
dns-server 192.168.11.194  
ip dhcp pool lan1v20  
network 192.168.10.0 255.255.255.128  
default-router 192.168.10.1  
dns-server 192.168.11.194  
ip dhcp pool lan2  
network 192.168.11.0 255.255.255.128  
default-router 192.168.11.1  
dns-server 192.168.11.194  
ip dhcp pool lan3  
network 192.168.11.128 255.255.255.192  
default-router 192.168.11.129  
dns-server 192.168.11.194  
!  
interface Serial0/2/0  
ip address 10.0.0.13 255.255.255.252  
ip nat inside  
!  
interface Serial0/2/1  
ip address 200.0.0.1 255.255.255.248  
ip nat outside  
!  
interface Serial0/3/0  
ip address 10.0.0.1 255.255.255.248  
ip nat inside  
!  
interface Serial0/3/1  
ip address 10.0.0.9 255.255.255.252  
ip nat inside  
!  
ip nat inside source list 1 interface Serial0/2/1 overload  
ip nat inside source static 192.168.11.194 200.0.0.3  
ip nat inside source static 192.168.11.195 200.0.0.4  
ip classless  
ip route 192.168.10.0 255.255.255.128 10.0.0.2  
ip route 192.168.10.128 255.255.255.128 10.0.0.2  
ip route 192.168.11.0 255.255.255.128 10.0.0.10  
ip route 192.168.11.128 255.255.255.192 10.0.0.14  
ip route 192.168.11.192 255.255.255.248 10.0.0.2  
!  
access-list 1 permit 192.168.10.0 0.0.0.255  
access-list 1 permit 192.168.11.0 0.0.0.127  
access-list 1 permit 192.168.11.128 0.0.0.63  
!  
end
```

Engineer Router

```
!  
version 15.4  
hostname Router  
!  
crypto isakmp policy 10  
  encr aes  
  authentication pre-share  
  group 2  
!  
crypto isakmp key cisco address 10.0.0.10  
!  
crypto ipsec transform-set ts esp-aes esp-sha-hmac  
!  
crypto map VPN 10 ipsec-isakmp  
  set peer 10.0.0.10  
  set transform-set ts  
  match address 110  
!  
interface GigabitEthernet0/0/0  
  no ip address  
  ip helper-address 10.0.0.1  
  ip nat inside  
  duplex auto  
  speed auto  
!  
interface GigabitEthernet0/0/0.10  
  encapsulation dot1Q 10  
  ip address 192.168.10.129 255.255.255.128  
  ip helper-address 10.0.0.1  
  ip nat inside  
!  
interface GigabitEthernet0/0/0.20  
  encapsulation dot1Q 20  
  ip address 192.168.10.1 255.255.255.128  
  ip helper-address 10.0.0.1  
  ip nat inside  
!  
interface GigabitEthernet0/0/0.30  
  encapsulation dot1Q 30  
  ip address 192.168.11.193 255.255.255.248  
  ip nat inside  
!  
interface Serial0/1/0  
  ip address 10.0.0.2 255.255.255.248  
  ip nat outside  
  clock rate 2000000  
  crypto map VPN  
!  
ip classless  
ip route 10.0.0.8 255.255.255.252 10.0.0.1  
ip route 10.0.0.12 255.255.255.252 10.0.0.1  
ip route 192.168.11.0 255.255.255.128 10.0.0.1  
ip route 192.168.11.128 255.255.255.192 10.0.0.1  
ip route 200.0.0.0 255.255.255.248 10.0.0.1  
!  
access-list 1 permit 192.168.10.0 0.0.0.255  
access-list 110 permit ip 192.168.11.192 0.0.0.7 192.168.11.0 0.0.0.127  
!  
end
```

Science Router

```

!
version 15.4
hostname Router
!
crypto isakmp policy 10
  encr aes
  authentication pre-share
  group 2
!
crypto isakmp key cisco address 10.0.0.2
!
crypto ipsec transform-set ts esp-aes esp-sha-hmac
!
crypto map VPN 10 ipsec-isakmp
  set peer 10.0.0.2
  set transform-set ts
  match address 110
!
interface GigabitEthernet0/0/0
  ip address 192.168.11.1 255.255.255.128
  ip helper-address 10.0.0.9
  ip nat inside
  duplex auto
  speed auto
!
interface Serial0/1/0
  ip address 10.0.0.10 255.255.255.252
  ip nat outside
  clock rate 2000000
  crypto map VPN
!
ip classless
ip route 10.0.0.0 255.255.255.248 10.0.0.9
ip route 10.0.0.12 255.255.255.252 10.0.0.9
ip route 192.168.10.0 255.255.255.128 10.0.0.9
ip route 192.168.10.128 255.255.255.128 10.0.0.9
ip route 192.168.11.128 255.255.255.192 10.0.0.9
ip route 200.0.0.0 255.255.255.248 10.0.0.9
ip route 192.168.11.192 255.255.255.248 10.0.0.9
!
access-list 1 permit 192.168.11.0 0.0.0.127
access-list 110 permit ip 192.168.11.0 0.0.0.127 192.168.11.192 0.0.0.7
!
end

```

Art Router

```

!
version 15.1
hostname Router
!
interface GigabitEthernet0/0
  ip address 192.168.11.129 255.255.255.192
  ip helper-address 10.0.0.13
  ip access-group 110 in
  ip nat inside
  duplex auto
  speed auto
!
interface Serial0/3/0
  ip address 10.0.0.14 255.255.255.252
  ip nat outside
  clock rate 2000000
!
router rip
!
ip classless
ip route 10.0.0.8 255.255.255.252 10.0.0.13
ip route 192.168.11.0 255.255.255.128 10.0.0.13
ip route 10.0.0.0 255.255.255.248 10.0.0.13
ip route 192.168.10.0 255.255.255.128 10.0.0.13
ip route 192.168.10.128 255.255.255.128 10.0.0.13
ip route 200.0.0.0 255.255.255.248 10.0.0.13
ip route 192.168.11.192 255.255.255.248 10.0.0.13
!
access-list 1 permit 192.168.11.128 0.0.0.63
access-list 110 deny icmp 192.168.11.128 0.0.0.63 192.168.11.192 0.0.0.7
access-list 110 permit ip any any
!
end

```

ISP Router

```
!  
version 15.1  
hostname ISP  
!  
interface Serial0/3/0  
  ip address 200.0.0.2 255.255.255.248  
  clock rate 2000000  
!  
ip classless  
!  
end
```

Engineer Switches

Switch Destination Eng

```
!  
hostname Switch_denstination_eng  
!  
interface FastEthernet0/1  
  switchport mode trunk  
!  
interface FastEthernet0/2  
  switchport access vlan 10  
  switchport mode access  
!  
! (Interfaces Fa0/2 to Fa0/7 are in VLAN 10)  
! (Interfaces Fa0/8 to Fa0/20 are in VLAN 20)  
!  
interface FastEthernet0/8  
  switchport access vlan 20  
  switchport mode access  
!  
end
```

Switch Core Eng

```
!  
hostname Switch_core_eng  
!  
interface FastEthernet0/1  
  switchport mode trunk  
!  
interface FastEthernet0/2  
  switchport mode trunk  
!  
interface FastEthernet0/3  
  switchport mode trunk  
!  
end
```

Switch Access Eng

```
!  
hostname Switch_aceess_eng  
!  
interface FastEthernet0/1  
  switchport trunk allowed vlan 30  
  switchport mode trunk  
!  
interface FastEthernet0/2  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/3  
  switchport access vlan 30  
  switchport mode access  
!  
end
```

(Note: Other intermediate switches like Switch3, Switch4, Switch5, and Switch6 are basic L2 pass-through devices with Fa0/1 and Fa0/2 configured as switchport mode trunk without any specific VLAN access assignment).